

Boktiar Ahmed Bappy



Professional Summary

Data Scientist with 5+ years of experience in Generative AI, LLMOps, and Autonomous AI Systems. Experience in designing, deploying, and scaling enterprise-grade AI solutions across healthcare, self-development, and automation. Skilled in multi-agent architectures, RAG pipelines, and model optimization, delivering measurable impact on scalability, accuracy, and adoption. Recognized contributor at Omdena for global AI collaborations. Winner of the APICTA Merit Award 2021 and ICT Award 2020.

Technical Skills

- Programming & AI:** Python, sklearn, PyTorch, TensorFlow, Keras, Hugging Face, LangChain, LlamaIndex, RAG, Agentic AI, Crew AI, LangGraph
- MLOps & Cloud:** MLflow, SageMaker, Kubernetes, Docker, Git, DVC, CICD, AWS, GCP, Azure
- Databases:** SQL, NoSQL, Vector Databases (Pinecone, Faiss, Weaviate, Chroma DB).
- Specialties:** Generative AI, RAG, Multi-Agent Systems, LLM Fine-Tuning, Computer Vision, NLP

Experience

AI/ML Engineer (Generative AI)

Dallas, Texas, United States

Quantum Human

05/2025 – Present

- Contributed to Sentara Development:** Built and iteratively refined healthcare AI solutions, integrating LLMs and Generative AI for remote patient assistance and wellness tracking using LangChain, Agentic RAG, Pinecone, FastAPI.
- Biometrics Data Integration:** Integrated the Oura Ring and the EEG Device in Sentara to get real-time users biometrics data using the Oura and the EEG Device API
- Memory Integration:** Architected and developed an advanced layered memory system for long-term contextual understanding and personalization.
- Deployment:** Deployed Sentara as CI/CD on Render Cloud

Associate Manager & Senior Data Scientist

Bengaluru, Karnataka. India

Euron

10/2024 – 04/2025

- Contributed to Euri Assistant Development:** Designed and implemented a GenAI-powered support assistant, reducing query response and resolution times by 70% for Euron’s user base.
- Multi-LLM Integration:** Integrated multiple large language models (GPT, LLaMA, Gemini, Qwen, DeepSeek, Mistral, and other open-source models) as selectable options for users, while extending model performance with an enhanced knowledge base.
- Tool & Agentic Pipeline Development:** Built an agentic orchestration pipeline enabling seamless integration of diverse tools to enhance the Euri Assistant’s capabilities.
- AI Leadership & Innovation:** Directed AI research and product development across Computer Vision, Natural Language Processing, and Generative AI, providing mentorship and driving innovation initiatives.

Associate Manager & Data Scientist

Bengaluru, Karnataka. India

iNeuron.ai

01/2022 - 07/2024

- Medical Chatbot Development:** Designed and implemented a GenAI-powered medical assistant capable of delivering context-aware health guidance, supporting patient interactions, and integrating with healthcare knowledge bases to ensure accuracy, reliability, and safety.
- Advanced RAG Architectures:** Engineered an Advanced RAG with LangChain, Pinecone, Llama, and a Knowledge base to enable autonomous decision-making and AI-driven workflows.
- Deployment:** Deployed the chatbot using CI/CD pipeline on AWS Cloud
- Computer Vision Platform:** Delivered scalable vision solutions integrated into iNeuron Intelligence and PW Skills platforms.

Projects

- Enhancing Regulatory Decision-Making through a Retrieval-Augmented Generation (RAG):** The project focused on digitizing physical records including board papers, meeting minutes, and trade agreements—into a secure and easily accessible digital archive, ensuring the preservation and quick retrieval of historical documents. On top of this foundation, a Retrieval-Augmented Generation (RAG) based Large Language Model (LLM) was developed to provide users with a chat-based Q&A system for seamless access to relevant records. This innovation empowers regulatory officers to make more informed, timely, and accurate decisions by reducing delays in accessing historical data, ultimately enhancing policy enforcement and regulatory updates.
- ImageSeeker - An Auto Image Classification Library:** ImageSeeker is a Python library designed to streamline image classification tasks by reducing the amount of code developers need to write, allowing them to focus on solving the most critical parts of their problem. The package was developed with TensorFlow and Keras at its core, leveraging Keras’s principle of progressive disclosure of complexity to provide both simplicity for quick workflows and flexibility for more advanced use cases. To ensure accessibility, the library was published on PyPI, enabling seamless installation via pip.
- Vision-Based Attendance System:** The Vision-Based Attendance System was developed to automate and streamline attendance tracking using computer vision techniques. By leveraging facial recognition and deep learning models, the system accurately identifies individuals in real time, eliminating the need for manual entry or physical authentication methods. This solution not only improves efficiency and accuracy but also enhances security, making it highly suitable for educational institutions, workplaces, and large-scale events where reliable attendance management is essential.

Education

- Bachelor's in Computer Science & Engineering – Royal University of Dhaka**
- Diploma in Electronics Engineering – Jhenaidah Polytechnic Institute**

Research & Publication

- Development of Multiple Combined Regression Methods for Rainfall Measurement**